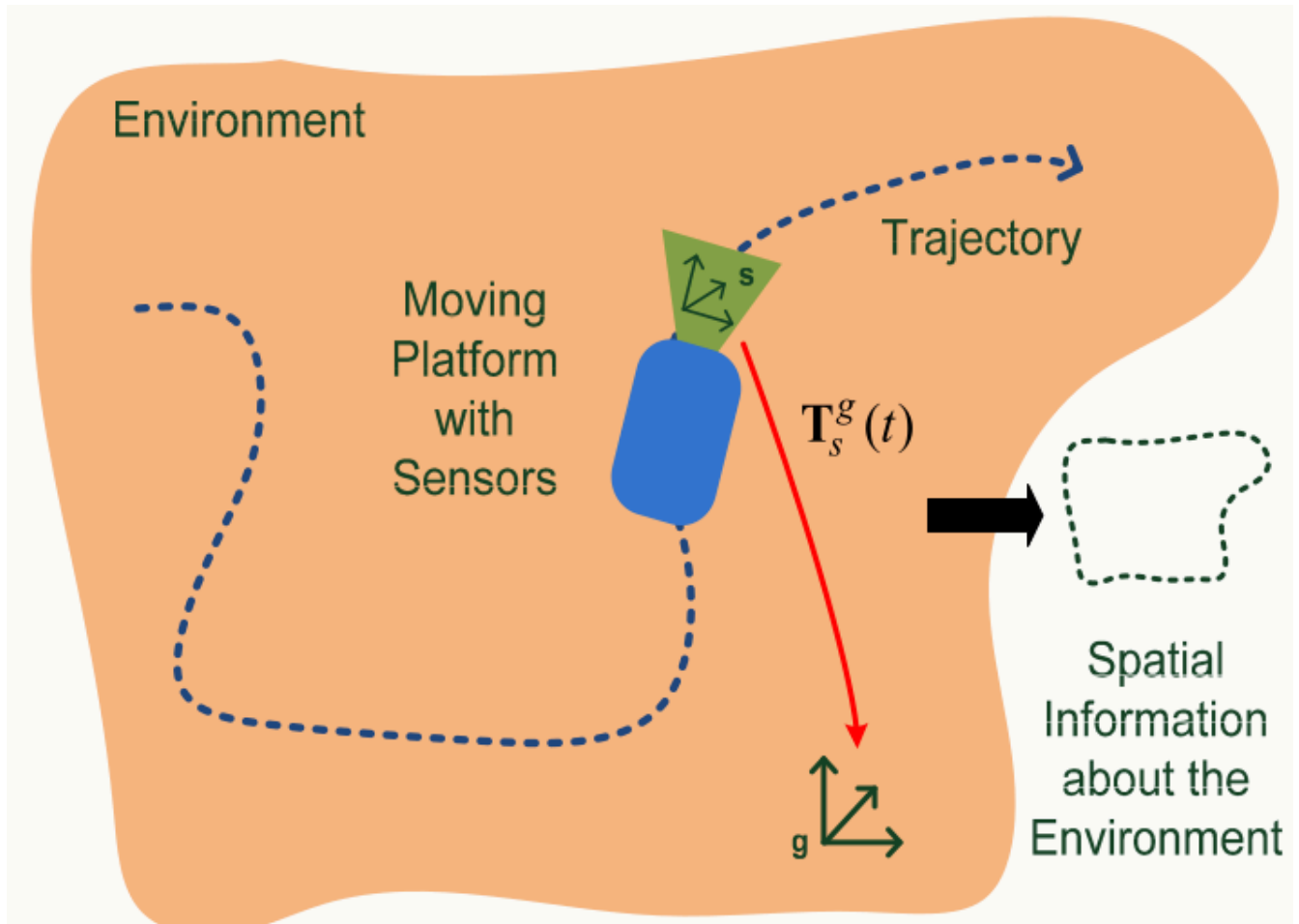


L4: Mobile Laser Scanning

- Mobile Mapping



Sensor Synchronisation

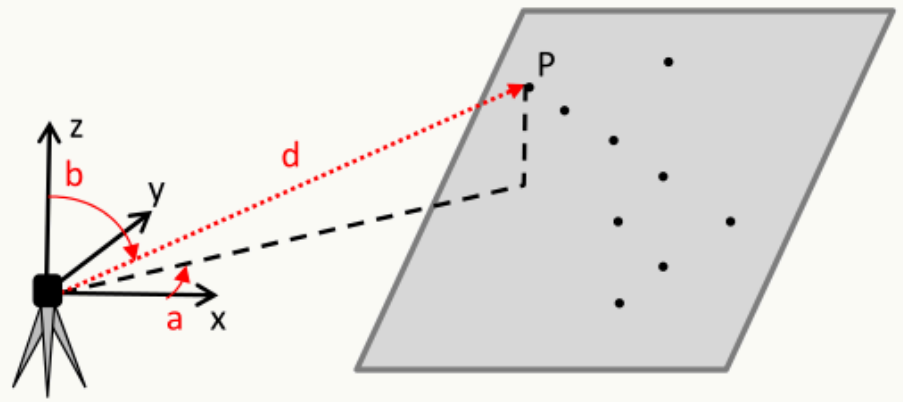
$$\begin{aligned}
 \mathbf{p}_{object}^{global}(t_s) &= \mathbf{T}_{sensor}^{global}(t_s) \cdot \mathbf{p}_{object}^{sensor}(t_s) \\
 &= \mathbf{T}_{body}^{global}(t_s) \cdot \mathbf{T}_{sensor}^{body} \cdot \mathbf{p}_{object}^{sensor}(t_s)
 \end{aligned}$$

$\underbrace{\hspace{10em}}$	$\underbrace{\hspace{10em}}$	$\underbrace{\hspace{10em}}$	$\underbrace{\hspace{10em}}$
object coordinate In the global coordinate system	position und orientierung of the vehicle in the global coordinate system	System calibration	object coordinate in the sensor (e.g. scanner) coordinate system

- Usage of frame and coordinate system equivalent in this lecture

Sensor Frame - Laser Scanner

- Fixed to sensor
- Observations given

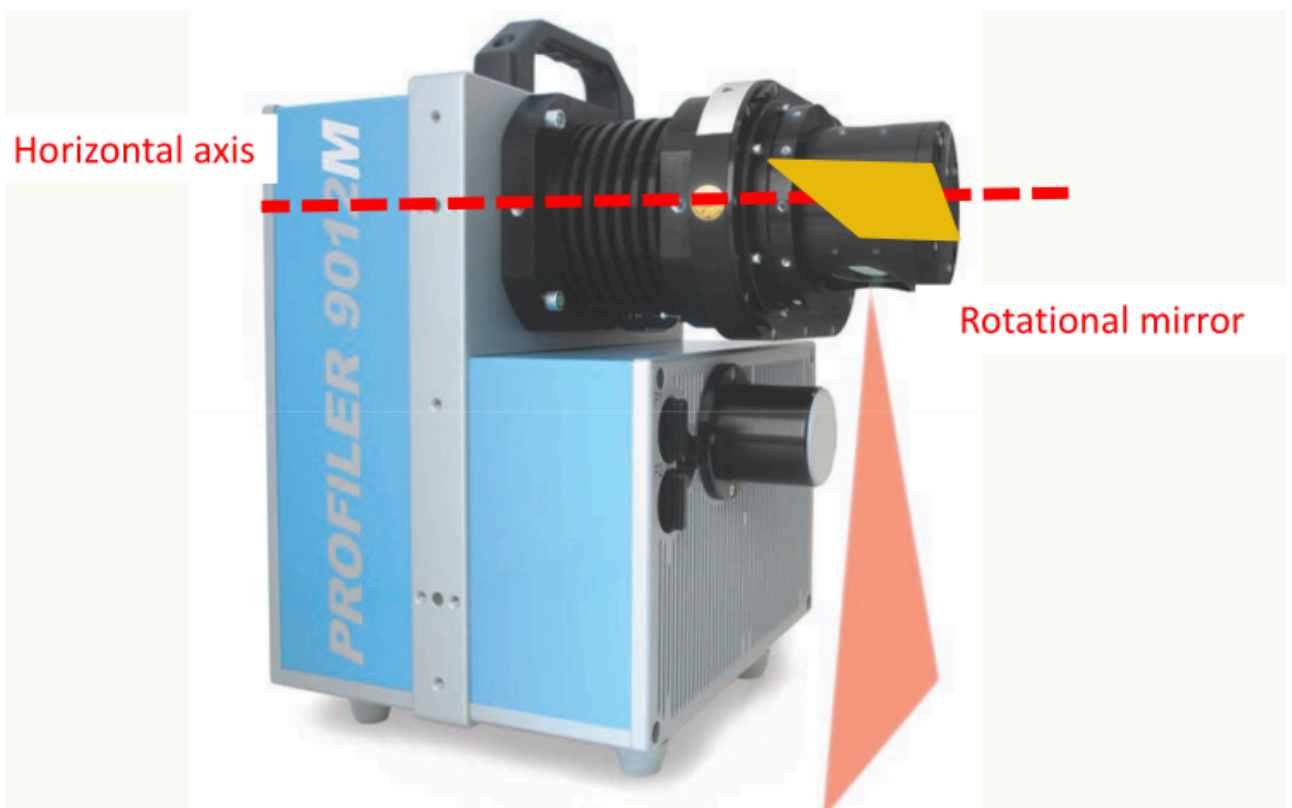


$$\begin{bmatrix} x_s \\ y_s \\ z_s \end{bmatrix} = \begin{bmatrix} d \cdot \sin b \cdot \sin a \\ d \cdot \sin b \cdot \cos a \\ d \cdot \cos b \end{bmatrix} \longleftrightarrow \begin{bmatrix} d \\ b \\ a \end{bmatrix} = \begin{bmatrix} \sqrt{x_s^2 + y_s^2 + z_s^2} \\ \arccos(z_s/d) \\ \arctan(y_s/x_s) \end{bmatrix}$$

Cartesian Coordinates

Polar Coordinates

- Profile Scanner



- No rotation around the vertical axis ($a=0$)

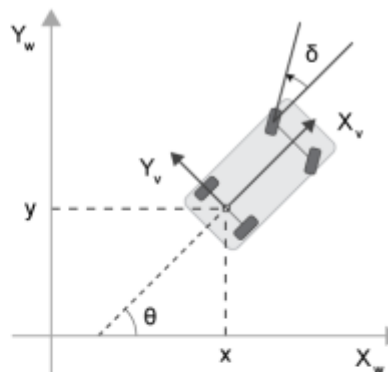
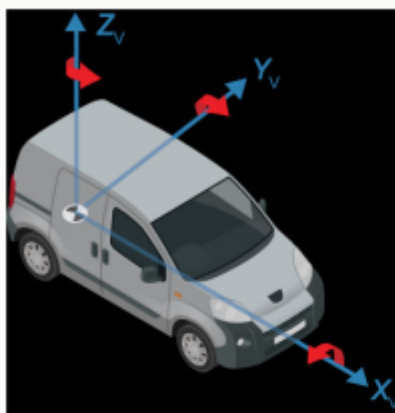
-

$$\begin{bmatrix} x_s \\ y_s \\ z_s \end{bmatrix} = \begin{bmatrix} d \cdot \sin b \cdot \sin a \\ d \cdot \sin b \cdot \cos a \\ d \cdot \cos b \end{bmatrix} = \begin{bmatrix} 0 \\ d \cdot \sin b \\ d \cdot \cos b \end{bmatrix}$$

$$\begin{bmatrix} x_e \\ y_e \\ z_e \end{bmatrix} = \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} + \mathbf{R}_n^e(L, B) \mathbf{R}_b^n(\phi, \theta, \psi) \cdot \begin{bmatrix} \Delta x \\ \Delta y \\ \Delta z \end{bmatrix} + \mathbf{R}_s^b(\alpha, \beta, \gamma) \cdot \begin{bmatrix} 0 \\ d \cdot \sin b \\ d \cdot \cos b \end{bmatrix}$$

Body/Vehicle/Platform Frame

- Frame which is fixed to the moving platform
- Origin and axis known in some "world frame"
- Usually body frame not same as sensor frame

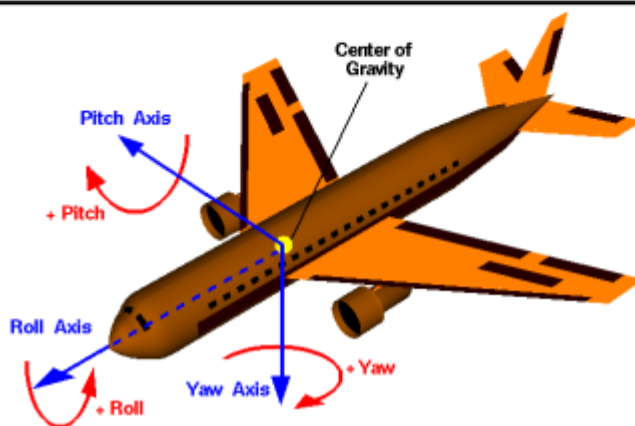


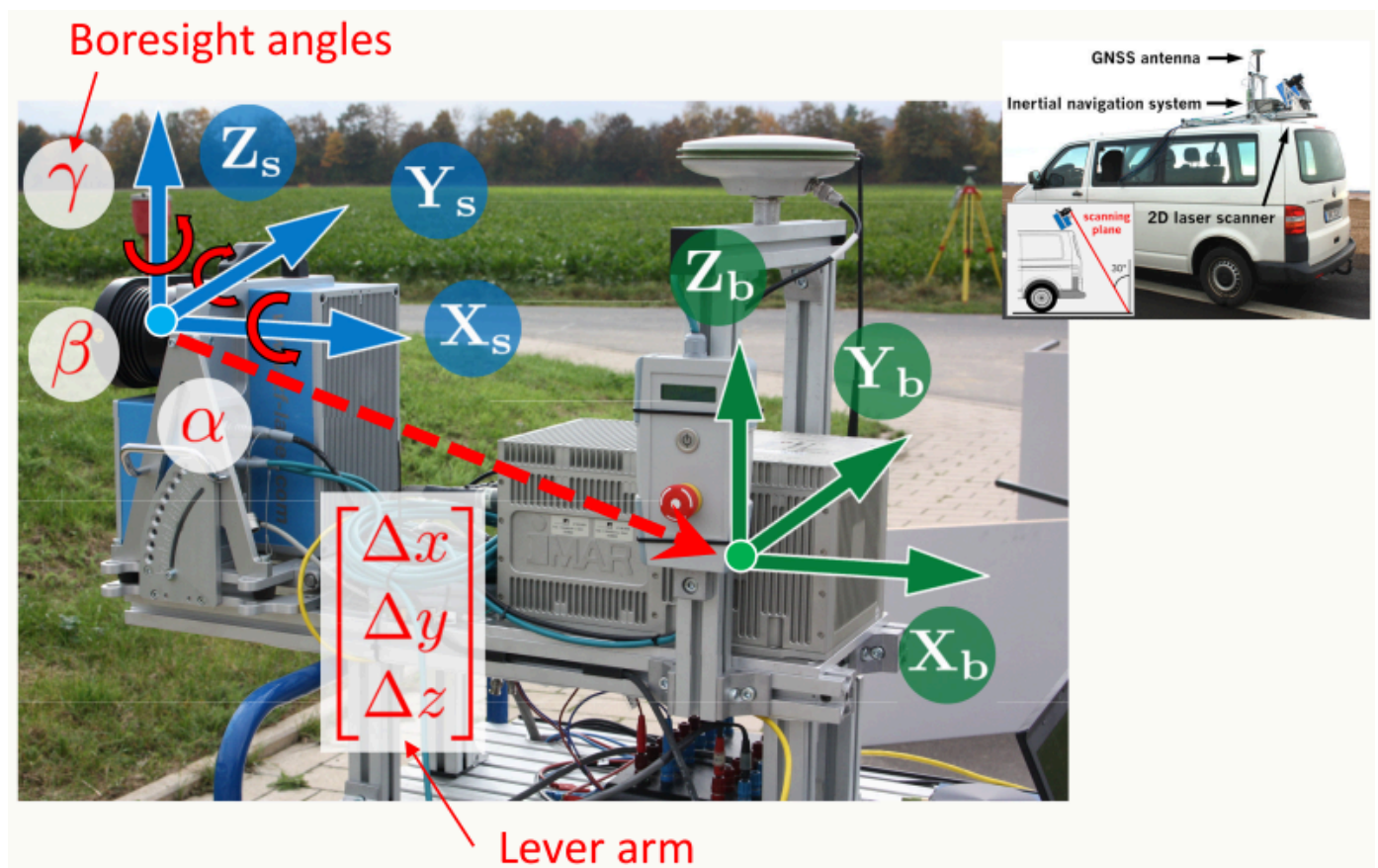
Matlab Autonomous Driving Toolbox



Aircraft Rotations Body Axes

Glenn
Research
Center





System Calibration

- Determination of transformation (lever arm and boresight angles)
 - sensor frame -> body frame

$$\begin{bmatrix} x_e \\ y_e \\ z_e \end{bmatrix} = \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} + \mathbf{R}_n^e(L, B) \mathbf{R}_b^n(\phi, \theta, \psi) \cdot \begin{bmatrix} \Delta x \\ \Delta y \\ \Delta z \end{bmatrix} + \mathbf{R}_s^b(\alpha, \beta, \gamma) \cdot \begin{bmatrix} 0 \\ d \cdot \sin b \\ d \cdot \cos b \end{bmatrix}$$

$$\mathbf{R}_s^b(\alpha, \beta, \gamma) = \begin{bmatrix} \cos(\beta) \cdot \cos(\gamma) & \cos(\alpha) \cdot \sin(\gamma) + \cos(\gamma) \cdot \sin(\alpha) \cdot \sin(\beta) & \sin(\alpha) \cdot \sin(\gamma) - \cos(\alpha) \cdot \cos(\gamma) \cdot \sin(\beta) \\ -\cos(\beta) \cdot \sin(\gamma) & \cos(\alpha) \cdot \cos(\gamma) - \sin(\alpha) \cdot \sin(\beta) \cdot \sin(\gamma) & \cos(\gamma) \cdot \sin(\alpha) + \cos(\alpha) \cdot \sin(\beta) \cdot \sin(\gamma) \\ \sin(\beta) & -\cos(\beta) \cdot \sin(\alpha) & \cos(\alpha) \cdot \cos(\beta) \end{bmatrix}$$

Body Frame -> Global Frame

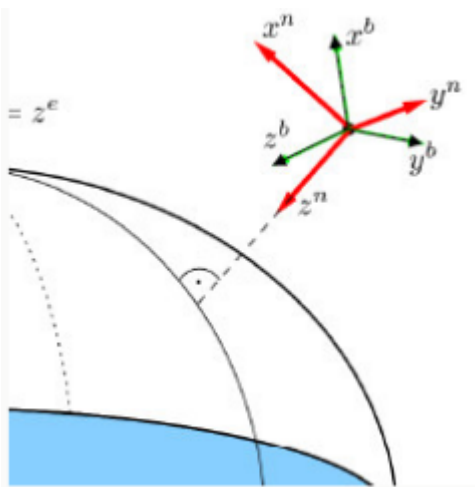
- Trajectory
- Transformations
 - Body moving w.r.t global frame
 - transformation parameters are changing
 - Trajectory estimation

Navigation Frame

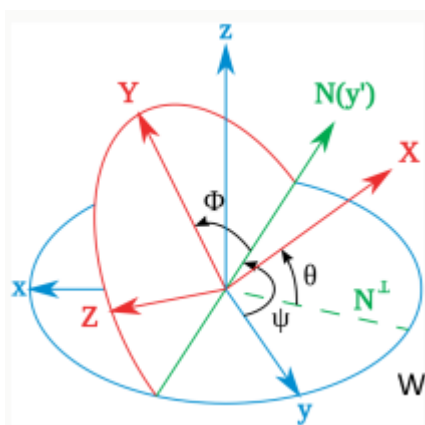
$$\begin{bmatrix} x_e \\ y_e \\ z_e \end{bmatrix} = \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} + \mathbf{R}_n^e(L, B) \mathbf{R}_b^n(\phi, \theta, \psi) \begin{bmatrix} x_b \\ y_b \\ z_b \end{bmatrix}$$

Rotation from body to navigation frame, parametrized by 3 angles

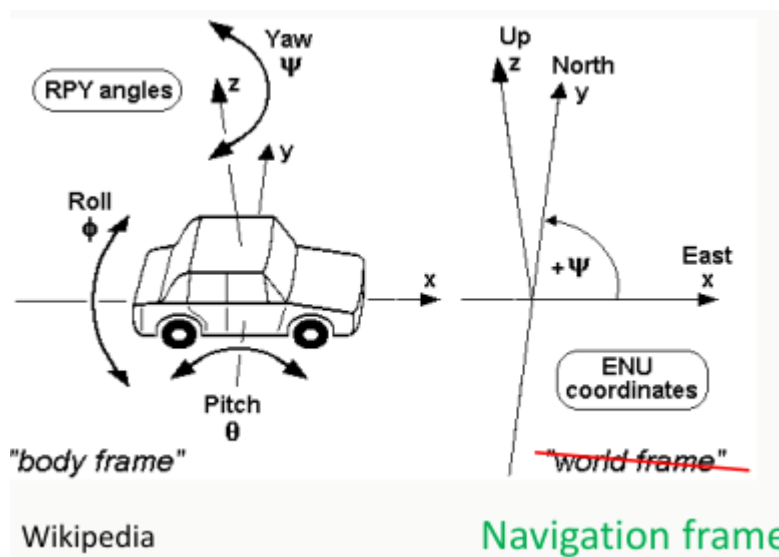
- **Navigation Frame:**
 - Origin: origin of body frame
 - Axis: North-East-Down (NED) or East-North-Up (ENU)
- Usually reference frame for altitudes



- Euler angles
 - Roll: ϕ
 - Pitch: θ
 - Yaw: ψ or heading



- Ground Based Vehicles
 - n-frame: ENU
 - b-frame: forward-left-up
 - ϕ = Yaw = 0° -> Heading towards East

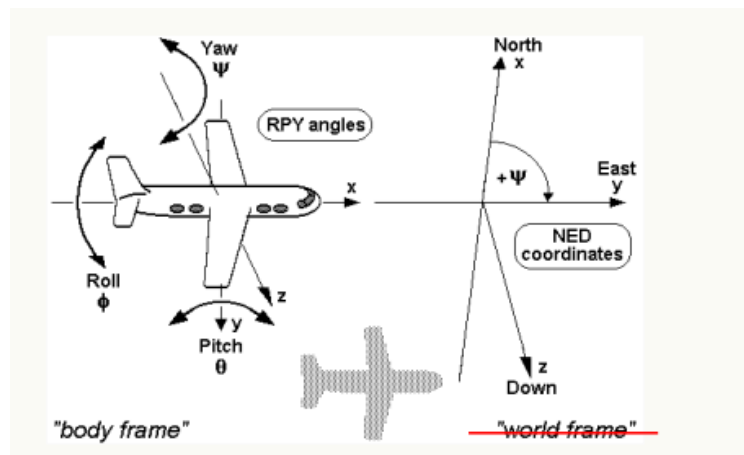


◦ Wikipedia

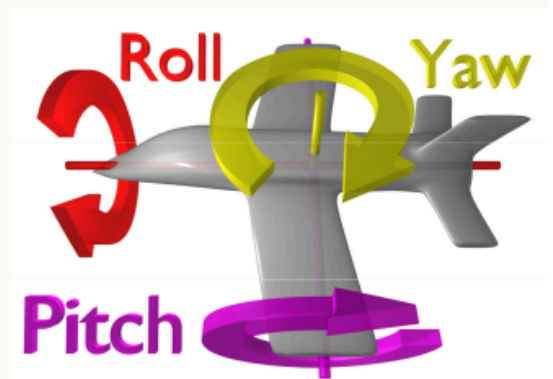
Navigation frame

- Aerial Vehicles

- n-frame: NED
- b-frame: forward-right-down
- yaw: $0^\circ \rightarrow$ Heading towards North



◦



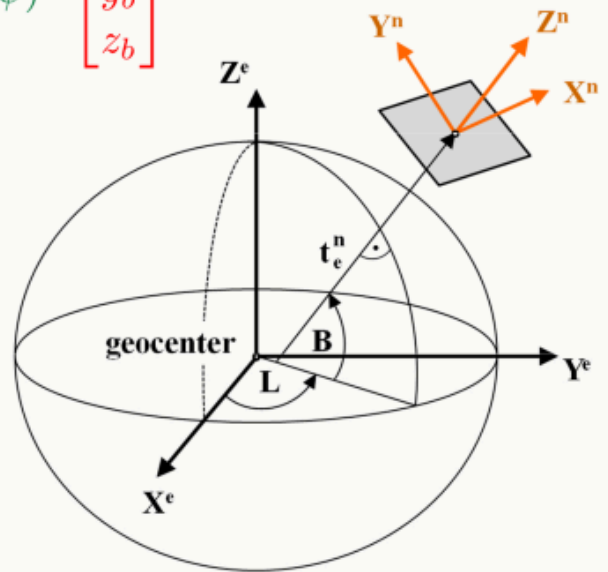
Navigation Frame to Earth Frame

$$\begin{bmatrix} x_e \\ y_e \\ z_e \end{bmatrix} = \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} + \mathbf{R}_n^e(L, B) \mathbf{R}_b^n(\phi, \theta, \psi) \cdot \begin{bmatrix} x_b \\ y_b \\ z_b \end{bmatrix}$$

$$= \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} + \mathbf{R}_n^e(L, B) \cdot \begin{bmatrix} x_n \\ y_n \\ z_n \end{bmatrix}$$

Latitude L

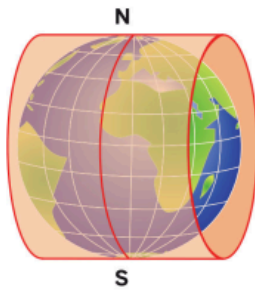
Longitude B



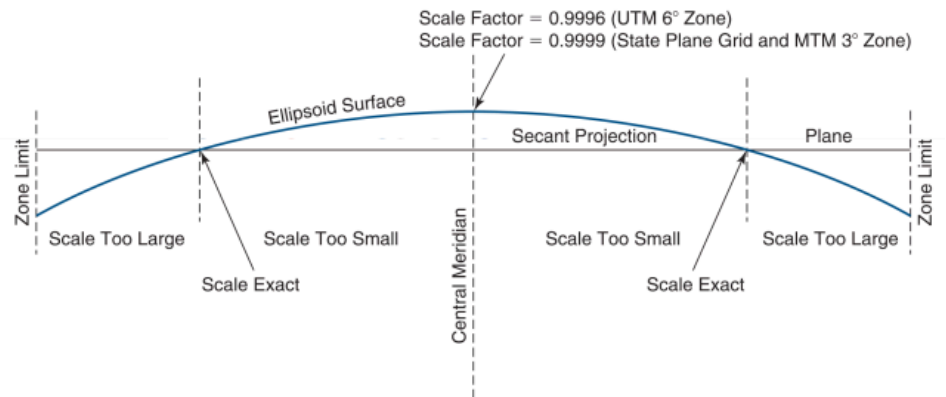
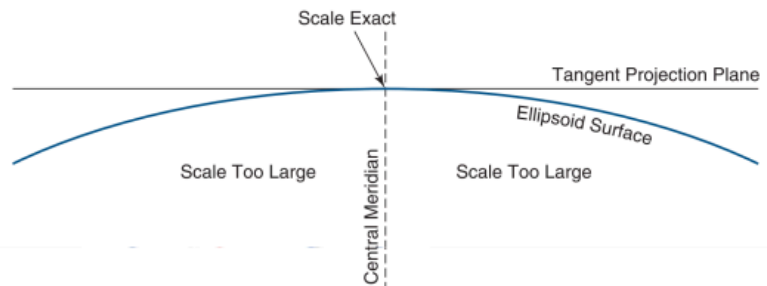
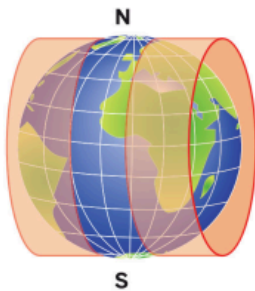
$$\mathbf{R}_n^e = \begin{bmatrix} -\sin(L) & -\sin(B) \cdot \cos(L) & \cos(B) \cdot \cos(L) \\ \cos(L) & -\sin(B) \cdot \sin(L) & \cos(B) \cdot \sin(L) \\ 0 & \cos(B) & \sin(B) \end{bmatrix}$$

Does not need to be calculated every time within a local area. Only Every 100-200m

- Earth Frame Coordinates
 - ECEF: Earth Centred Earth Fixed
 - X, Y, Z
 - -> Cartesian, but not with reference to Earth surface
 - -> good for calculation
 - Map + heights
 - Map projection necessary



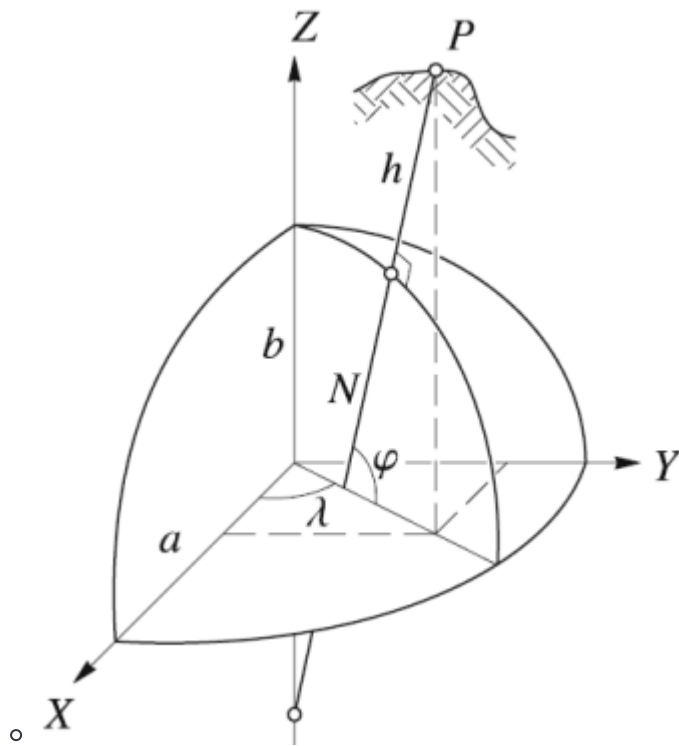
Above: Gauß-Krüger => tangential
Below: Universal transverse mercator (UTM)
=> intersectional



-
- UTM: 6° wide zones
- Scale factor at central meridian 0.9996
- Distortion

Offset to central meridian [km]	Distortion in distance [mm] (distance 100 m, height = 0 m above GRS80 ellipsoid)
0	- 40
100	- 28
180	0
200	+ 9

- Geodetic/ellipsoidal
 - B, L, h
 - -> Useful on a global scale
 - Reference surface is ellipsoid
 - not cartesian
 - Need of information about reference ellipsoid
 - a, 1/f



- Ellipsoidal coordinates are easier to interpret on earth
- Ellipsoidal coordinates need definition of reference ellipsoid (e.g. GRS80, WGS84, Bessel)

• $\lambda, \varphi, h \Rightarrow X, Y, Z$

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} (N(\varphi) + h) \cdot \cos \varphi \cdot \cos \lambda \\ (N(\varphi) + h) \cdot \cos \varphi \cdot \sin \lambda \\ \left(\frac{b^2}{a^2} N(\varphi) + h \right) \cdot \sin \varphi \end{bmatrix}$$

$$N(\varphi) = \frac{a^2}{\sqrt{a^2 \cdot \cos^2 \varphi + b^2 \cdot \sin^2 \varphi}}$$

... transverse radius of curvature

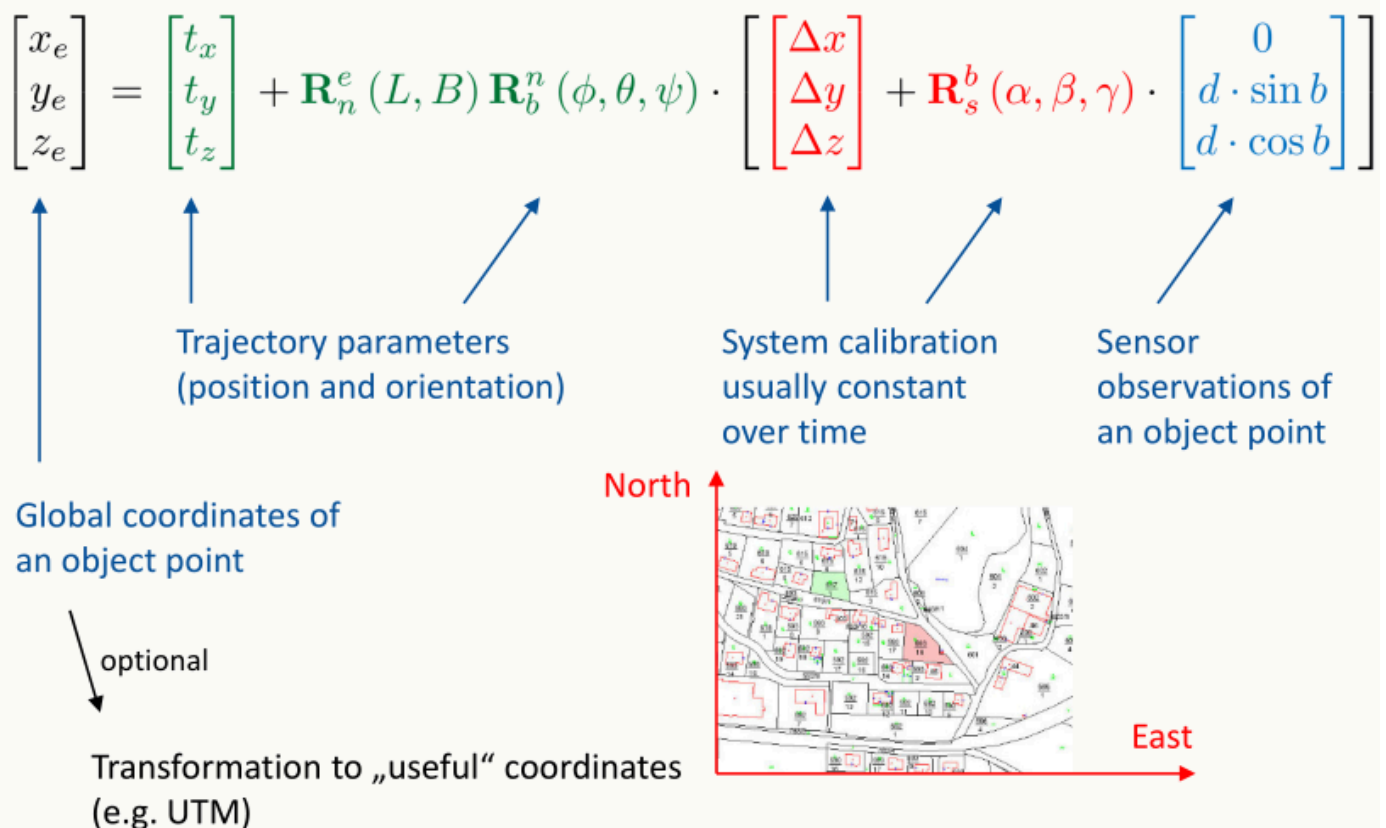
- $X, Y, Z \Rightarrow \lambda, \varphi, h$

$$\begin{bmatrix} \varphi \\ \lambda \\ h \end{bmatrix} = \begin{bmatrix} \tan^{-1} \left(\frac{Z}{\sqrt{X^2 + Y^2}} \cdot \left(1 - \frac{a^2 - b^2}{a^2} \cdot \frac{N(\varphi)}{N(\varphi) + h} \right)^{-1} \right) \\ \tan^{-1} \left(\frac{Y}{X} \right) \\ \frac{\sqrt{X^2 + Y^2}}{\cos \varphi} - N(\varphi) \end{bmatrix}$$

- Iterate: $N(\varphi) = \frac{a^2}{\sqrt{a^2 \cdot \cos^2 \varphi + b^2 \cdot \sin^2 \varphi}}$

o

WRAP UP – GEOREFERENCING EQUATION



- Trajectory Data given in UTM Coordinates
 - o Navigation Frame have same axis direction as global frame
 - Global Frame serves as reference frame for roll, pitch, yaw
 - Navigation to Earth Rotation is not necessary
- Distance measurements of the scanner has to be scaled along East component (according to the distance to the central meridian)